

Govind Singh Tanwar

+91-7828087932 | govindsingh7126@gmail.com | linkedin.com/in/govindsi-v | github.com/govind-sing

Forward Deployed Engineer — TypeScript · Node.js · React · REST APIs · AWS · FastAPI

TECHNICAL SKILLS

Languages & Scripting: TypeScript, JavaScript (ES6+), Python, Bash, C++, SQL
Frontend: React.js, Next.js (App Router, Server Actions, Server Components), Tailwind CSS, WebSocket client
Backend & APIs: Node.js, Express.js, FastAPI, REST API design, JWT authentication, RBAC, custom middleware pipelines, WebSocket, async
AI & Integration: RAG pipelines, ChromaDB, Gemini embeddings, LLM-backed API design, local model inference, third-party API integration
Databases: PostgreSQL, MongoDB, Redis, SQL schema design, data modeling, referential integrity, Supabase
Infrastructure & DevOps: AWS (EC2, Lambda, API Gateway), Linux, Docker, Docker Compose, nginx, GitHub Actions (CI/CD), Git, Vercel

PROJECTS

- DayForge** | *Next.js, FastAPI, PostgreSQL, ChromaDB, Docker, AWS EC2* | GitHub | GitHub 2026
- Independently designed and shipped a full stack AI application end-to-end — Next.js frontend, 2 decoupled FastAPI backends exposing 15+ REST endpoints, and a WebSocket layer for real-time agent communication — containerized via Docker Compose and deployed to production on AWS EC2 using Bash deploy scripts and nginx reverse proxy.
 - Built a RAG pipeline (ChromaDB + Gemini embeddings, 100+ vectors) powering a conversational AI agent with DB-persisted memory — retrieving top-k relevant chunks via vector similarity to keep responses grounded in user data; architected an 11-table PostgreSQL schema with rolling 14-day adherence scoring.
 - Load-tested 4 authenticated endpoints at 100 concurrent users with 0% error rate; login at 514 req/s, p95 307ms; frontend scored 100/95/100/100 on Lighthouse — deployed and monitored entirely via Linux CLI on EC2.
- Debatify** | *React, Node.js, Express.js, MongoDB, AWS Lambda, JWT* | GitHub | GitHub 2025
- Built a full stack MERN social platform — React frontend with protected routes, role-gated UI, and real-time notification state, consuming a 47-route Node.js/Express REST API with JWT auth, email OTP, RBAC, and a layered custom middleware pipeline.
 - Designed a MongoDB schema across 8 collections handling nested content hierarchies, a social graph with follow relationships, and an internal event-driven notification system that fans out alerts on user interactions.
 - Deployed the REST API serverlessly on AWS Lambda via serverless-http — preserving the full Express middleware chain with zero refactoring; frontend on Vercel with environment-based API switching between local and production.
- SplitMint** | *Next.js 15, TypeScript, PostgreSQL, Supabase, GitHub Actions, Vercel* | GitHub | GitHub 2026
- Built a production-grade full stack expense splitter on Next.js 15 App Router with Server Actions and Server Components — eliminating a separate API layer and co-locating data fetching with UI for seamless user experience.
 - Designed a normalized PostgreSQL schema (groups, participants, expenses, splits, profiles) with cascading deletes and foreign-key constraints enforcing referential integrity at the database level — no orphaned records possible.
 - Implemented a greedy debt-settlement algorithm resolving multi-party balances in minimum transactions — validated against all combinatorial edge cases; CI/CD pipeline auto-deploys on every push with zero manual steps.

EDUCATION

Motilal Nehru National Institute of Technology Allahabad Prayagraj, UP
Bachelor of Technology — Biotechnology 2022 – 2026

- Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems

ACHIEVEMENTS

1st Place, Avishkar 2024 (MNNIT Allahabad Annual Technical Fest) — built and shipped a full-stack app for a live canteen startup in a timed team development event.

Competitive Programming: Codeforces Specialist (1400+), CodeChef 3★, 350+ LeetCode problems solved.